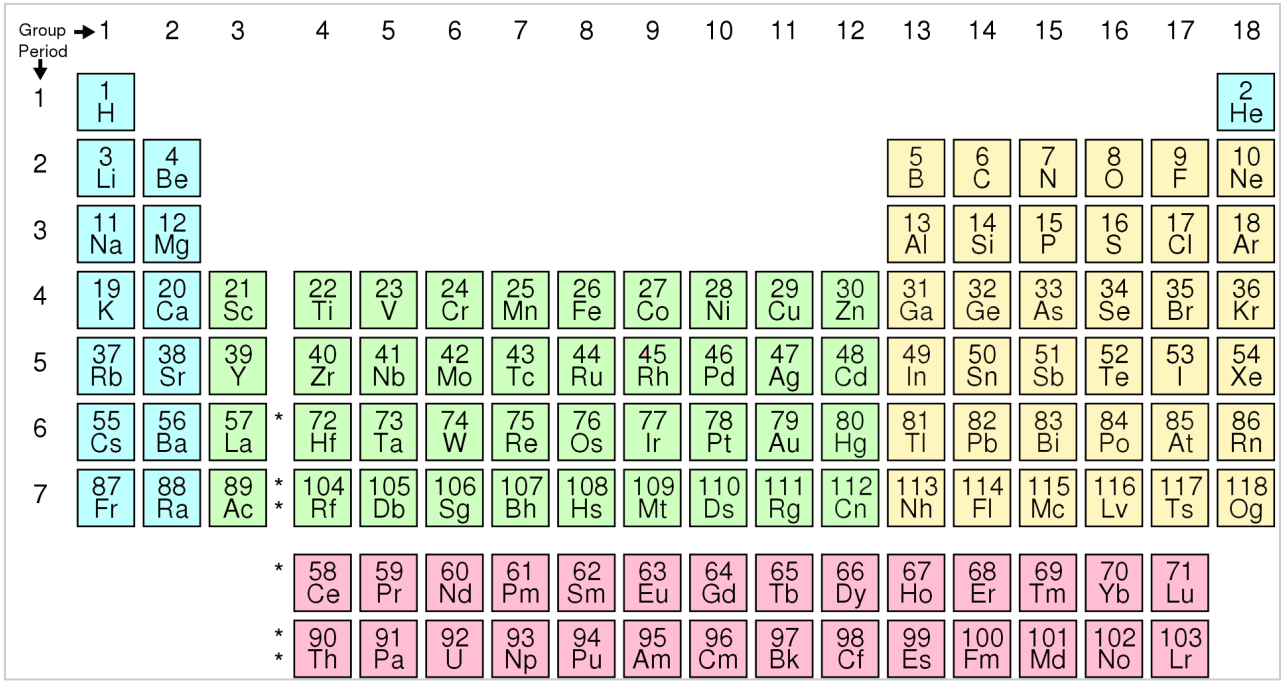
NTNU
Faculty of Information Technology
and Electrical Engineering
Department of Electronic Systems



# Exam H25 - Solutions

**1**




*Fill in the blanks in the following section of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)*

We can create mobile carriers in semiconductors by purposely introducing impurities into a crystal. This process is called doping.

We can create a p-doped piece of Ge by replacing some of the atoms in the crystal with

Select alternative (Si, B, P)-atoms. This type of doping is called Select alternative

(intrinsic doping, acceptor doping, donor doping)and will Select alternative (make the crystal negatively charged., not affect the net charge of the crystal., make the crystal positively charged.).

We can create an n-type Ge material by replacing some of the atoms in the crystal with

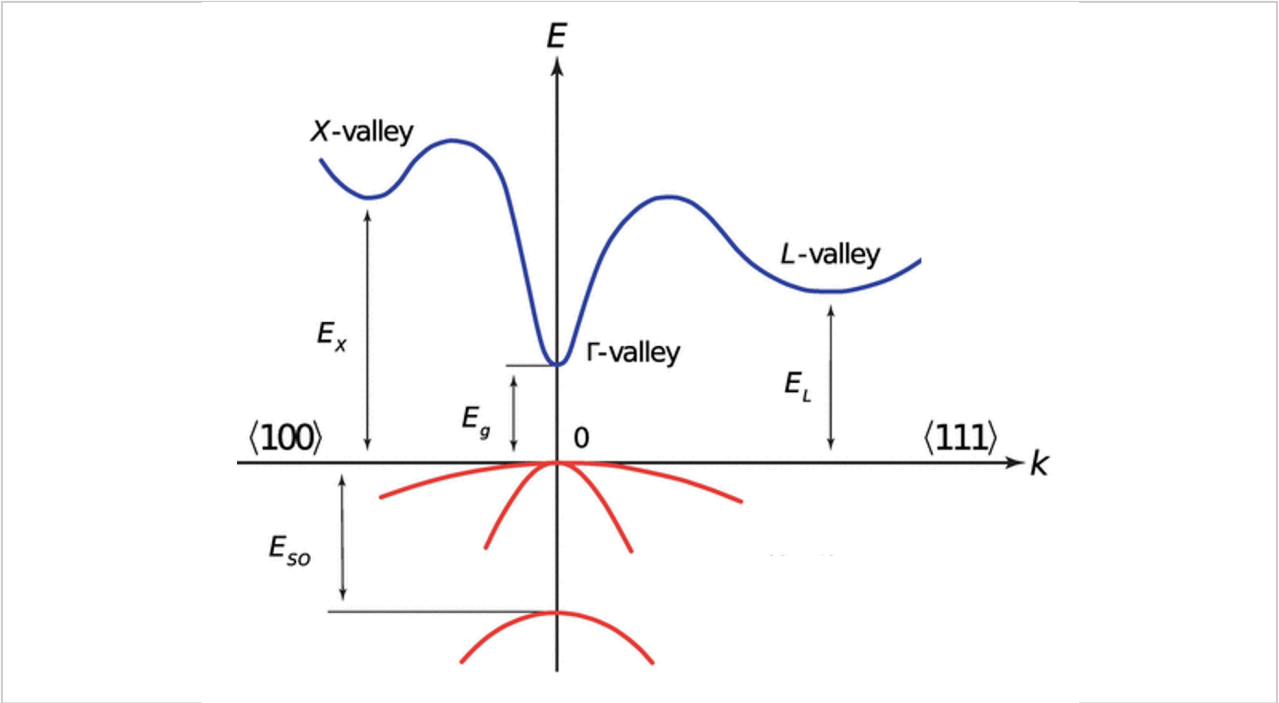
Select alternative (In, C, As)-atoms. This type of doping will introduce additional

Select alternative (holes, electrons) in the crystal and Select alternative (increase the mobility of the electrons in the system., increase the mobility of the holes in the system, increase the Fermi energy in the system., decrease the Fermi energy in the system.)

---

Maximum marks: 6

**2**




The figure shows the band structure of GaAs. In which energy valley of the GaAs conduction band will an electron have the highest mobility?

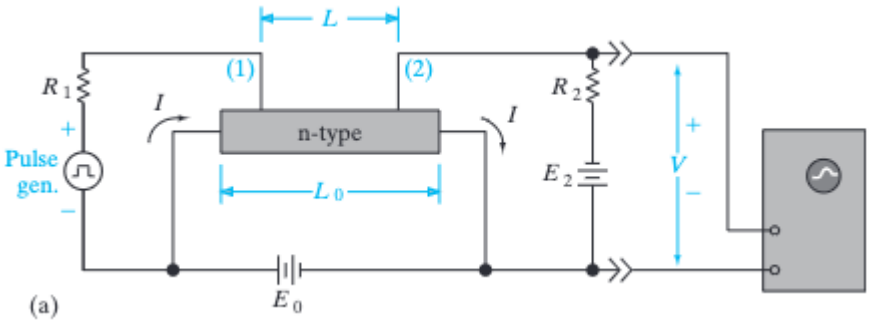
*(correct: 2p, wrong: -0p, blank: 0p, minimum score: 0p)*

**Select one alternative.**

- The $X$-valley
- The $L$-valley
- The $\Gamma$-valley ✔
- It will be the same in the entire band.

Maximum marks: 2

**3**




The figure above shows the experimental setup for the Haynes-Shockley experiment.
**Which parameters can be measured using the Haynes-Shockley experiment?**
*(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)*

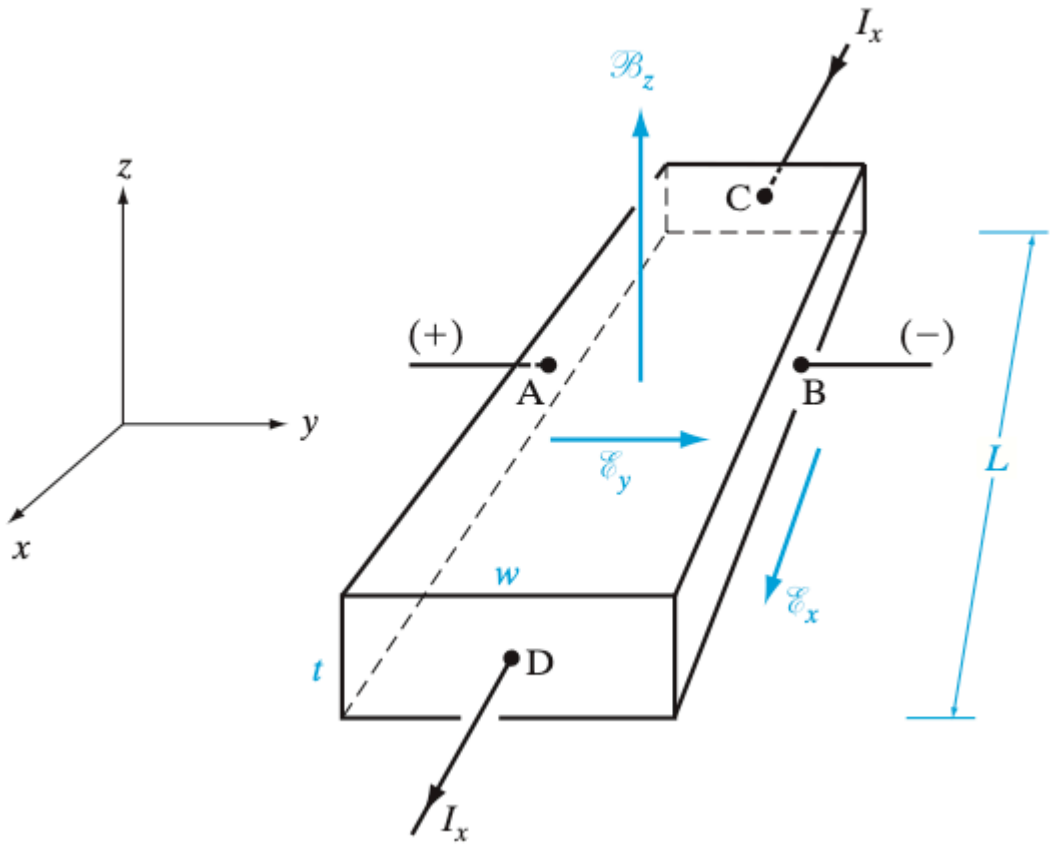
**Select one or more alternatives:**

- [ ] minority carrier concentration
- [ ] diffusion coefficient for minority carriers ✔
- [ ] mobility of majority carriers
- [ ] mobility of minority carriers ✔
- [ ] diffusion coefficient for majority carriers
- [ ] majority carrier concentration

---

Maximum marks: 4

**4**




The figure above shows the setup for the Hall experiment. The n-type semiconductor bar has a width $w = 0.01\text{cm}$, thickness $t = 10^{-3}\text{cm}$ and length $L = 0.5\text{cm}$. A magnetic field of $B_z = 10^{-4}\text{Wb/cm}^2$ is applied in the direction shown. For a current $I_x = 1\text{mA}$ we have $V_{AB} = -3\text{mV}$ and $V_{CD} = 100\text{mV}$.

Given: The force on a particle with charge $q$ and velocity $\vec{v}$, from a magnetic field $\vec{B}$, is $\vec{F} = q\vec{E} + q\vec{v} \times \vec{B}$

**Find the concentration and mobility of the majority carrier and fill in the numerical answer below.**

Majority carrier concentration : [ ] (1.9 - 2.2) $\cdot 10^{17}\text{cm}^{-3}$

*(correct: 4p, wrong: 0p, blank: 0p, minimum score 0p)*

Majority carrier mobility: [ ] (14000 - 16000) $\text{cm}^2/(\text{V}\cdot\text{s})$

*(correct: 4p, wrong: 0p, blank: 0p, minimum score 0p)*

*You may attach calculations if you wish.*

---

Maximum marks: 8

# Solution to Problem 4

**a)**
In the Hall-setup, the magnetic field will deflect charge carriers traveling along the $x$-axis according to the Lorentz force. Since $\boldsymbol{B} = B_z\hat{z}$ and the current is along the $x$-axis, this force is directed along the $y$-axis. This causes a charge imbalance, which gives rise to an electric potential that counteracts the magnetic field-induced deflection. When the two forces are equal, the accumulation of charge imbalance stops. Then, we have

$$\mathcal{E}_y = -v_x B_z, \quad \mathcal{E}_y = -\frac{V_{AB}}{w}. \tag{1}$$

Using the definition of the current density $J_x wt = I_x$, we also have we have

$$\frac{I_x}{wt} = J_x = qv_x N_{\text{carrier}} \quad \Longrightarrow \quad v_x = \frac{I_x}{qwtN_{\text{carrier}}}, \tag{2}$$

which gives

$$\frac{I_x B_z}{qwtN_{\text{carrier}}} = \frac{V_{AB}}{w}. \tag{3}$$

By rearranging the equation and assuming that the current is only due to majority charge carriers, this gives the majority carrier concentration

$$N_{\text{carrier}} = \frac{I_x B_z}{qtV_{AB}} = \frac{1\,\text{mA} \times 1 \cdot 10^{-4}\,\text{Wb/cm}^2}{1.6 \cdot 10^{-19}\,\text{A s} \times 1 \cdot 10^{-3}\,\text{cm} \times 3\,\text{mV}} \approx 2.081 \cdot 10^{17}\,\text{cm}^{-3} \tag{4}$$

**b)** From the formula sheet (3-43), we find that

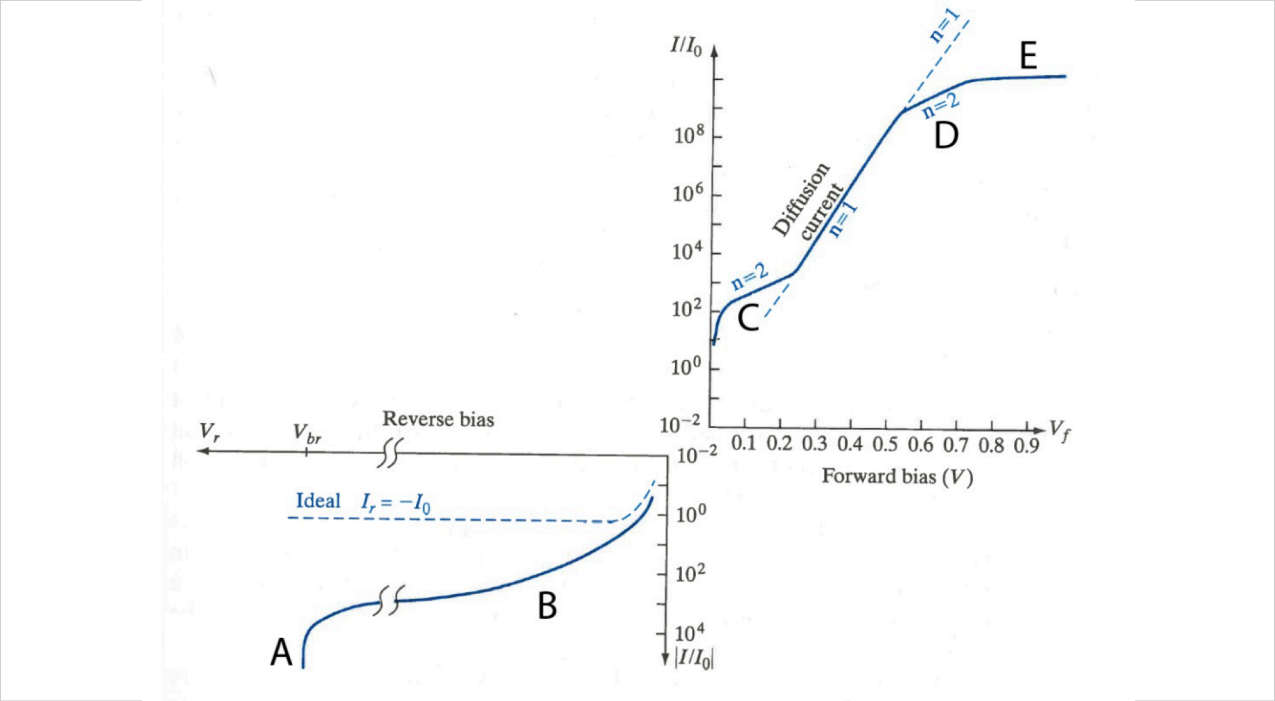
$$\frac{I_x}{wt} = q(n\mu_n + p\mu_p)\mathcal{E}_x. \tag{5}$$

Rearranging, assuming again that the majority carrier concentration is much larger than the minority carrier concentration, and using $V_{CD} = L\mathcal{E}_x$, we get

$$\begin{aligned}\mu_{\text{carrier}} &= \frac{I_x L}{wtqN_{\text{carrier}}V_{CD}} \\ &= \frac{1\,\text{mA} \times 5 \cdot 10^{-1}\,\text{cm}}{1.6 \cdot 10^{-19}\,\text{A s} \times 10^{-3}\text{cm} \times 10^{-2}\text{cm} \times 2.081 \cdot 10^{17}\,\text{cm}^{-3} \times 100\,\text{mV}} \\ &\approx 15 \cdot 10^3\,\text{cm}^2/\text{Vs}\end{aligned} \tag{6}$$

Note that it is not necessary to know the majority carrier type to calculate these numbers if you let $q$ be the charge of the given carrier type, but note that $V_{AB} < 0$ implies that the carriers are negatively charged, such that this is an n-doped semiconductor.

**5**




The illustration above highlights the most important differences in the I-V characteristics of an ideal pn-junction and a real pn junction.

*Fill in the blanks in the following section of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)*

In an ideal pn-junction the reverse saturation current is small and Select alternative (diffusion dominated, grows quadratically with the voltage, proportional to the voltage, voltage independent). For a real pn-junction this is not the case. In a real junction, for a sufficiently large reverse bias

reverse breakdown occurs (A). In a lightly doped junctions Select alternative (Shockley, Zener, Schottky, avalanche) breakdown is expected. This breakdown mechanism is characterized by

Select alternative (impact ionization, quantum mechanical tunneling, Eddy currents,

electrostatic discharge) due to a Select alternative (short diffusion length, excess of carriers,

large E-field, short length) in the Select alternative (contact potential, transition point,

depletion region, neutral p side). Furthermore, due to Select alternative (scattering, lack of carriers, electron-hole-pair generation, quantum mechanical tunneling) in the depletion region, the reverse current will grow (B) also before breakdown is reached.

In forward bias, the current will grow exponentially for both the ideal and the real pn-junction for moderate forward bias. Although the initial growth (C) will be a little slower for the real junction due

to Select alternative (an activation potential at the junction, high resistance in the depletion region, recombination in the depletion region, reduced mobility). For a sufficiently large forward

bias (D), the growth in forward current in the real junction will decrease as Select alternative (an abrupt junction, low-level injection, perfect bands, negligible drift) is no longer a good

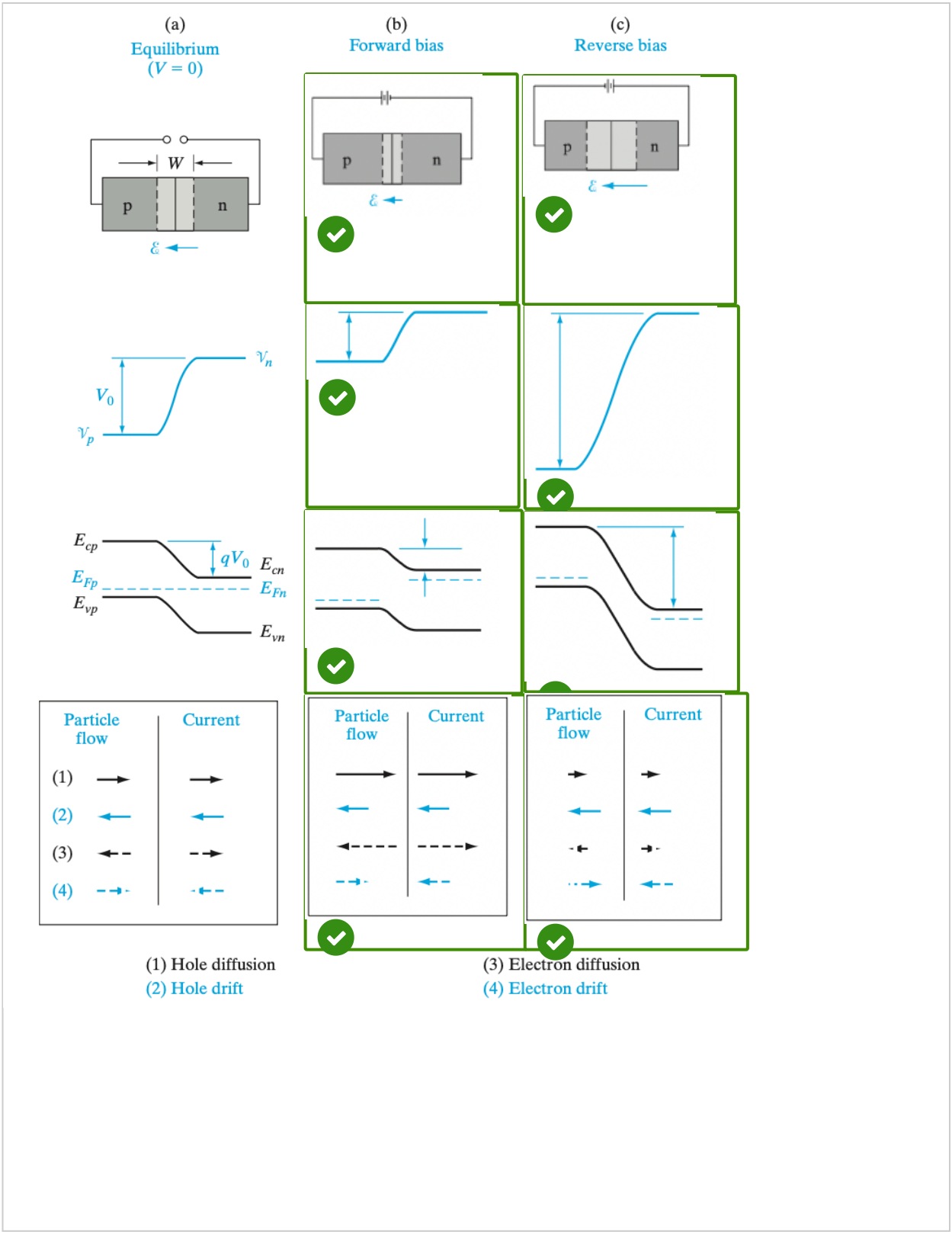
approximation and finally become linear (E) as Select alternative (ohmic effects, saturation effects, minority effects, thermal effects) becomes dominant.

---

Maximum marks: 9

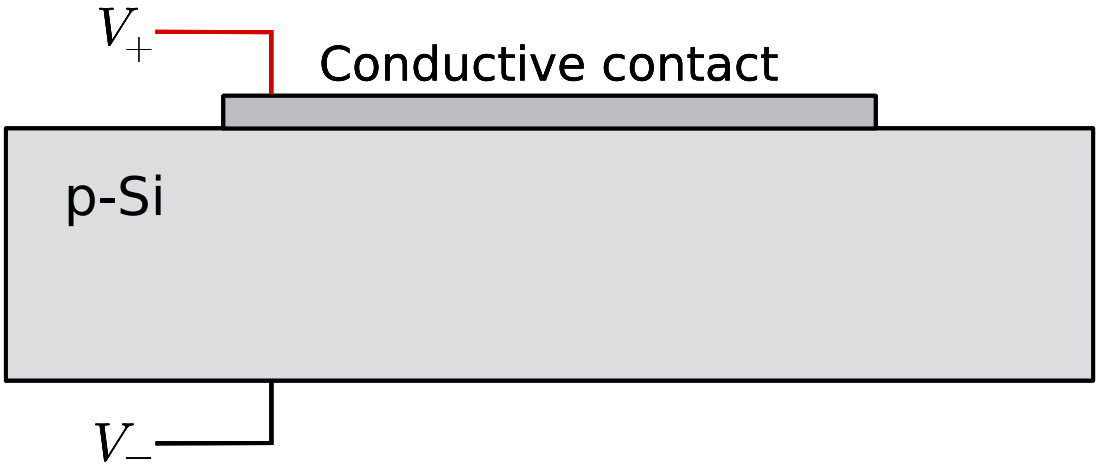
# Biasing in pn junctions

*Place the images in the correct positions. (correct: 0.5p, wrong: -1p, blank: -0p, minimum score: 0p)*




Correct. 4 of 4 marks. Try again

**7**




Anders is a researcher developing an electronic device at NTNU Nanolab. He wants to use a very thin conductive material as a contact on top of a p-doped Si substrate.

Being a careful researcher, he tests the resistance of the contacts before proceeding with his design by measuring the current after applying a small positive voltage over the contact and substrate.

To his surprise, he measures a *negative* current. Scratching his head, he measures the voltage over the contact and substrate in the configuration shown above, and finds a negative voltage of about 100mV. The magnitude of the voltage increases when he brings his work-lamp closer to the sample.[1]

*(correct: 2p, wrong: -1p, blank: -1p, minimum score 0p)*

**What type of contact has Anders made?**

Select alternative **(An ohmic contact, A rectifying contact, None of the above, A rectifying PN-junction).**

**What can we say about the work function $\Phi$ of the conductive contact material relative to the p-Si substrate?**

Select alternative (It is larger than the p-Si's work function, It is smaller than the p-Si's work function, We cannot say anything about the relative sizes of the work functions, It is equal to the p-Si's work function) **.**

**What sort of device has Anders accidentally made?**

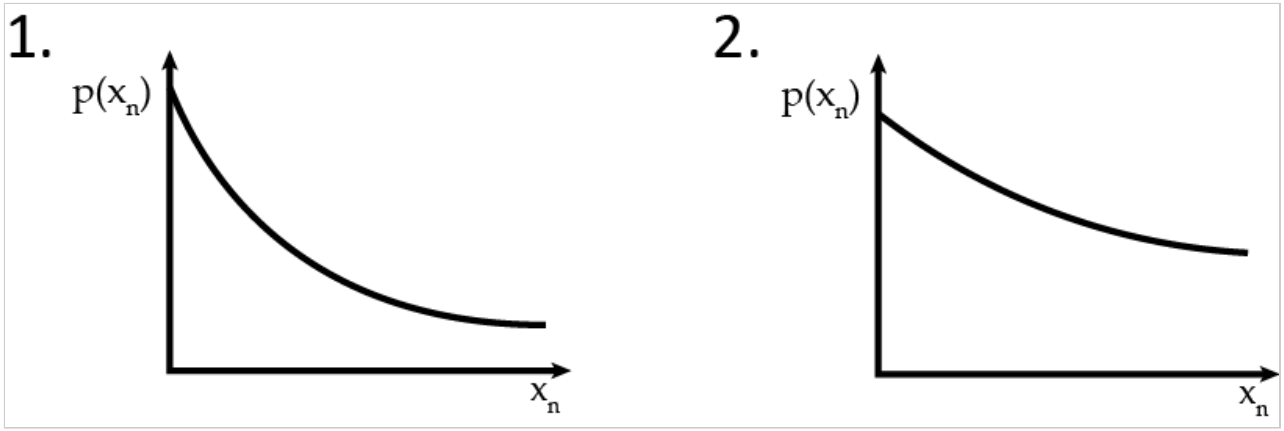
Select alternative **(A photodiode, A Schottky diode solar cell, A PN-diode solar cell, All of the above, An ohmic metal-semiconductor contact).**

[1] This is based on a true story from the lab.

---

Maximum marks: 6

**8**




The illustration shows the hole concentration on the n-side for two pn-junctions (device 1 and 2). Except for the doping levels we assume the two devices to be identical. Which of the following statements are correct ?

(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)

**Select one or more alternatives**

- [ ] In the figure we see a majority carrier distribution
- [ ] Both devices are currently in thermal equilibrium
- [ ] The n-side doping concentration is highest in device 1 ✔
- [ ] Both devices are currently under forward bias ✔
- [ ] The n-side doping concentration is highest in device 2
- [ ] The hole current accross the junction is highest in device 2
- [ ] In the figure we see a minority carrier distribution ✔

Maximum marks: 6

**9** An abrupt Si junction has the following parameters:
$N_a = 10^{17}\ cm^{-3}$
$N_d = 10^{15}\ cm^{-3}$
$\epsilon_r = 11.8$

**Calculate the Fermi level positions relative to the intrinsic Fermi level positions at 300 K in the p- and n-regions.**

Answers:

*(correct: 4p each, wrong: 0p, blank: 0p)*

In p-region: $E_{ip} - E_F =$ [ ] (0.39 - 0.42) eV

In n-region: $E_F - E_{in} =$ [ ] (0.27 - 0.30) eV

**Calculate the depletion region width, W.**

*(correct: 4p, wrong: 0p, blank: 0p)*

Answer: [ ] (0.945 - 0.96) $\mu m$

*You may attach calculations if you wish.*

---

Maximum marks: 12

# Solution to Problem 9

**a)** For both regions we may apply equation (3-25) from the formula sheet to get

$$N_\pm = n_i e^{\pm \frac{E_i - E_F}{k_B T}} \implies \pm(E_i - E_F) = k_B T \ln \frac{N_\pm}{n_i}, \tag{7}$$

where $\pm$ denotes the charge of the majority charge carrier. Thus, we have

$$(E_{ip} - E_F) = k_B T \ln \frac{N_a}{n_i} = 0.02585\,\text{eV} \ln \frac{10^{17}}{1.5 \cdot 10^{10}} \approx 0.406\,\text{eV} \tag{8}$$

$$(E_F - E_{in}) = k_B T \ln \frac{N_d}{n_i} = 0.02585\,\text{eV} \ln \frac{10^{15}}{1.5 \cdot 10^{10}} \approx 0.287\,\text{eV}. \tag{9}$$

**b)** Since we have $N_a \gg N_d$, we may approximate $N_a \gg N_d \implies N_a + N_d \approx N_a$ and use formula (5-57)

$$W = \sqrt{\frac{2\epsilon_0 \epsilon_r V_0}{q} \left( \frac{N_a + N_d}{N_a N_d} \right)} \approx \sqrt{\frac{2\epsilon_0 \epsilon_r V_0}{q N_d}}. \tag{10}$$

We need $V_0$, which is given by equation (5-8):

$$V_0 = \frac{k_B T}{q} \ln \frac{N_a N_b}{n_i^2} = 0.02585\,\text{V} \ln \frac{10^{17} 10^{15}}{(1.5 \cdot 10^{10})^2} \approx 0.69\,\text{V}. \tag{11}$$

In total, we thus have the sufficient answer

$$W \approx \sqrt{\frac{2 \times 11.8 \times 8.85 \cdot 10^{-12}\,\text{F/m} \times 0.69V}{1.6 \cdot 10^{-19}\,\text{A s} \times 10^{15}\text{cm}^{-3}}} \approx 0.951\,\mu\text{m}. \tag{12}$$

Not neglecting $N_d$, we see that $N_d = 10^{-2} N_a \implies \frac{N_a + N_d}{N_a N_d} = \frac{1.01 N_a}{N_a N_d} = \frac{1.01}{N_d}$, finally giving

$$W \approx \sqrt{\frac{2 \times 11.8 \times 8.85 \cdot 10^{-12}\,\text{F/m} \times 0.69V \times 1.01}{1.6 \cdot 10^{-19}\,\text{A s} \times 10^{15}\text{cm}^{-3}}} \approx 0.956\,\mu\text{m}. \tag{13}$$

**10** Consider a one-sided p+-n junction with a forward bias voltage $V_b >> \frac{kT}{q}$. What happens to the different current components if the bias voltage is increased by a factor 2?
*(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)*

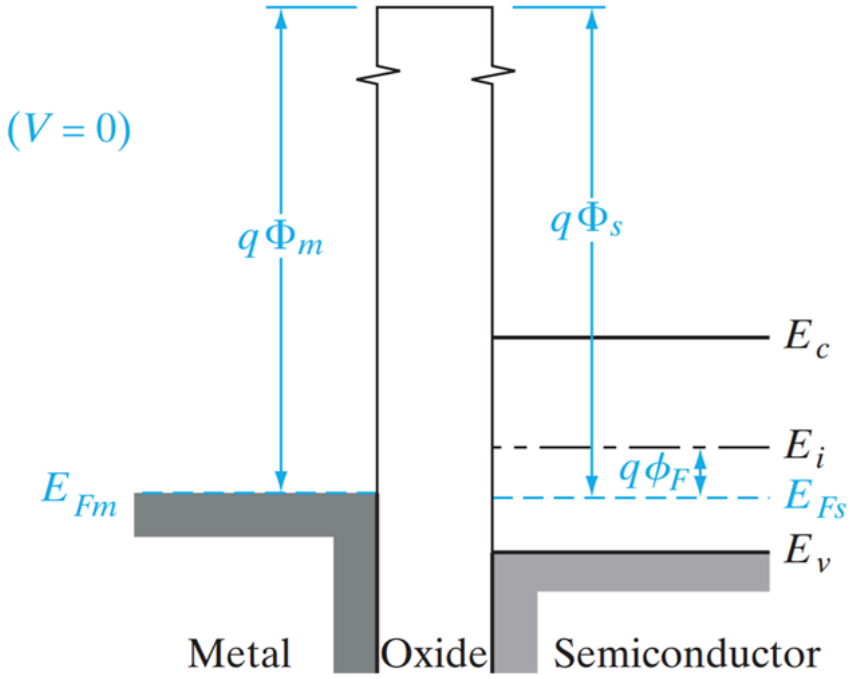
**Select one or more alternatives:**

- [ ] The diffusion of holes increase exponentially, the electron current remain unchanged
- [ ] The drift of holes and electrons remain approximately unchanged ✔
- [ ] The drift of holes and electrons increase by a factor 2
- [ ] The diffusion of holes increase exponentially and the diffusion of electrons decrease exponentially.
- [ ] The drift of holes increase by a factor 2 and the drift of electrons decrease by a factor 2.
- [ ] The diffusion of holes and electrons increase by a factor 2
- [ ] The diffusion of holes and electrons increase exponentially ✔

---

Maximum marks: 4

**11**




Consider a MOS capacitor consisting of a metal whose work function is equal to that of a channel region consisting of Si doped with $2 \times 10^{17} \mathrm{cm}^{-3}$ of Boron. Recall that the intrinsic carrier concentration of Si is approximately given by $n_i = 1.5 \times 10^{10}\ \mathrm{cm}^{-3}$. Assume $T = 300K$.

**a) Calculate the energy** $q\phi_F =$ [ ] (0.40 - 0.45) eV.

*(correct: 4p, wrong: 0p, blank: 0p, minimum score 0p)*

As the channel is brought into inversion, charges accumulate at the oxide-semiconductor interface. Si has a relative permittivity of $\epsilon_r = 11.8$

**b) Calculate the depletion charge of the channel at strong inversion**

$Q_d =$ [ ] (-250 - -230) $10^{-9}\ C/cm^2$

*(correct: 4p, wrong: 0p, blank: 0p, minimum score 0p)*

The gate oxide has a relative permittivity of $\epsilon_r = 3.9$ and has a thickness $d = 10\,\mathrm{nm}$. Assume no interface charges in the gate.

**c) Find the threshold voltage for this MOS-channel**

$V_T =$ [ ] (1.45 - 1.65) $V$

*(correct: 4p, wrong: 0p, blank: 0p, minimum score 0p)*

*You may attach calculations if you wish.*

---

Maximum marks: 12

# Solution to Problem 11

**a)** We recognize that calculating the quantity $q\phi_F$ for the Boron (acceptor) doped region equates to calculating the difference between the intrinsic and actual fermi energies of the semiconductor. This gives

$$q\phi_F = E_{ip} - E_F = k_B T \ln \frac{N_a}{n_i} = 0.02585\,\text{eV} \ln \frac{2 \cdot 10^{17}}{1.5 \cdot 10^{10}} \approx 0.424\,\text{eV}. \tag{14}$$

**b)** At strong inversion, we know that the semiconductor band bending at the interface, given by $\phi_s$ has been bent across the fermi level so strongly that the majority carrier concentration at the interface corresponds to that of a semiconductor with the opposite doping, i.e., $\phi_s = 2\phi_F$. We thus know the potential over the junction to calculate $W(\phi_s)$ as in problem 9. Each acceptor in the depletion region claims an electron of charge $-q$. This gives the total charge in the depletion region of the p-doped n-channel (formula 6-32)

$$Q_d = -qN_a W(2\phi_F) = -2\sqrt{\epsilon_r \epsilon_0 N_a q \phi_F} \tag{15}$$

Inserting the given quantities, we have

$$Q_d = -2\sqrt{11.8 \times 8.85 \cdot 10^{-12}\,\text{F/m} \times 2 \cdot 10^{17}\,\text{cm}^{-3} \times 424\,\text{meV}} \approx -238 \cdot 10^{-9}\,\text{C/cm}^{-2}. \tag{16}$$

**c)** Given the assumptions given by the problem, we have

$$V_T = -\frac{Q_d}{C_i} + 2\phi_F. \tag{17}$$

To find the threshold voltage, we therefore only need the gate capacitance, which is

$$C_i = \frac{\epsilon_{\text{oxide}} \epsilon_0}{d_{\text{oxide}}} = 3.9 \times \frac{8.85 \cdot 10^{-12}\,\text{F/m}}{10\,\text{nm}} \approx 345\,\text{nF/cm}^2 \tag{18}$$

Inserting this into the expression for the threshold voltage, we have

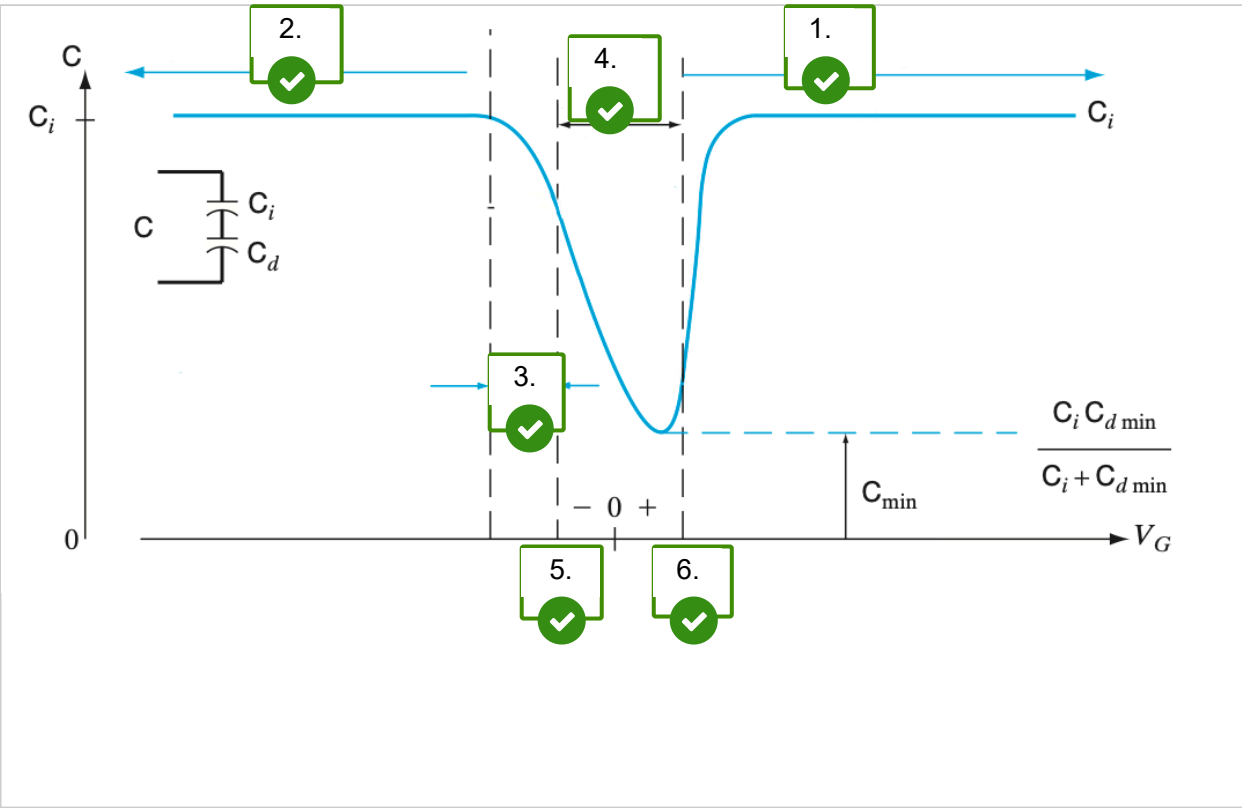
$$V_T = 0.708\text{V} + 0.848\text{V} \approx 1.56\,\text{V} \tag{19}$$

# MOS Capacitance

*Place the labels in the correct positions. (correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)*

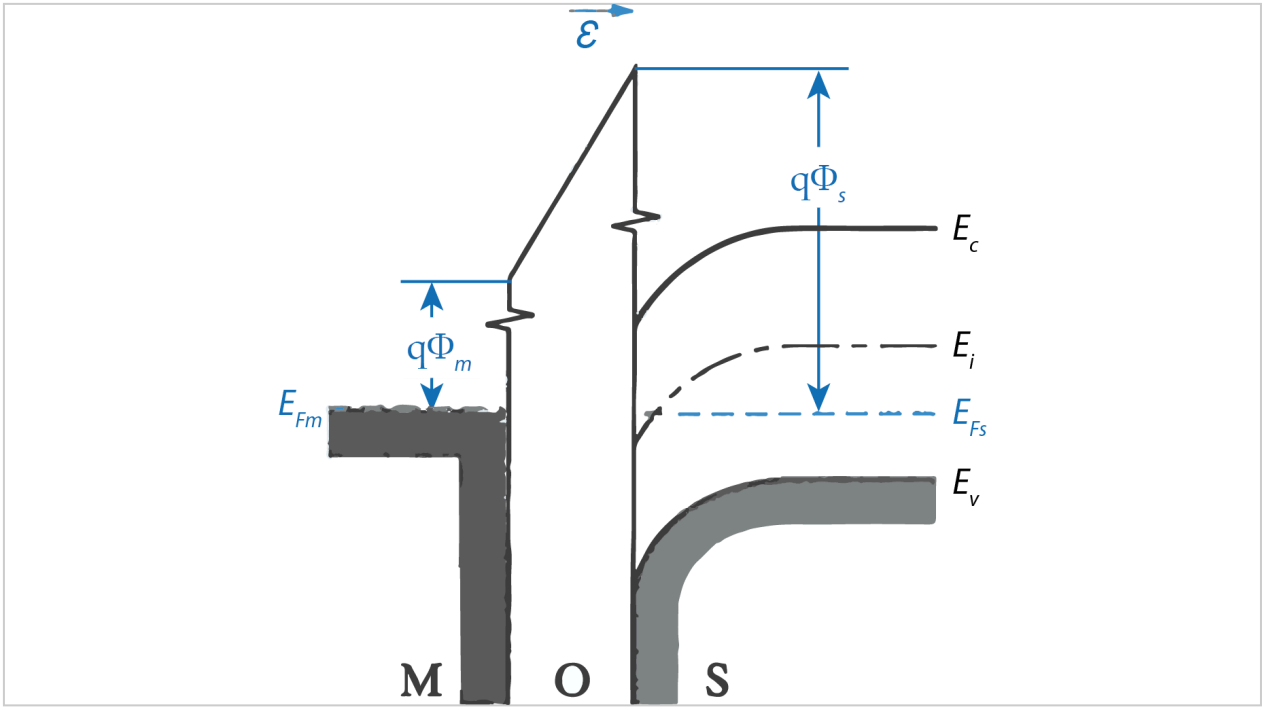
1. Strong inversion
2. Strong accumulation
3. Weak accumulation
4. Depletion and weak inversion
5. $V_{FB}$
6. $V_T$

**Place the labels in the correct positions:**




Correct. 6 of 6 marks. Try again

**13**




The illustration shows the band diagram for a certain MOS capacitor. Which of the following statements are correct for this device?

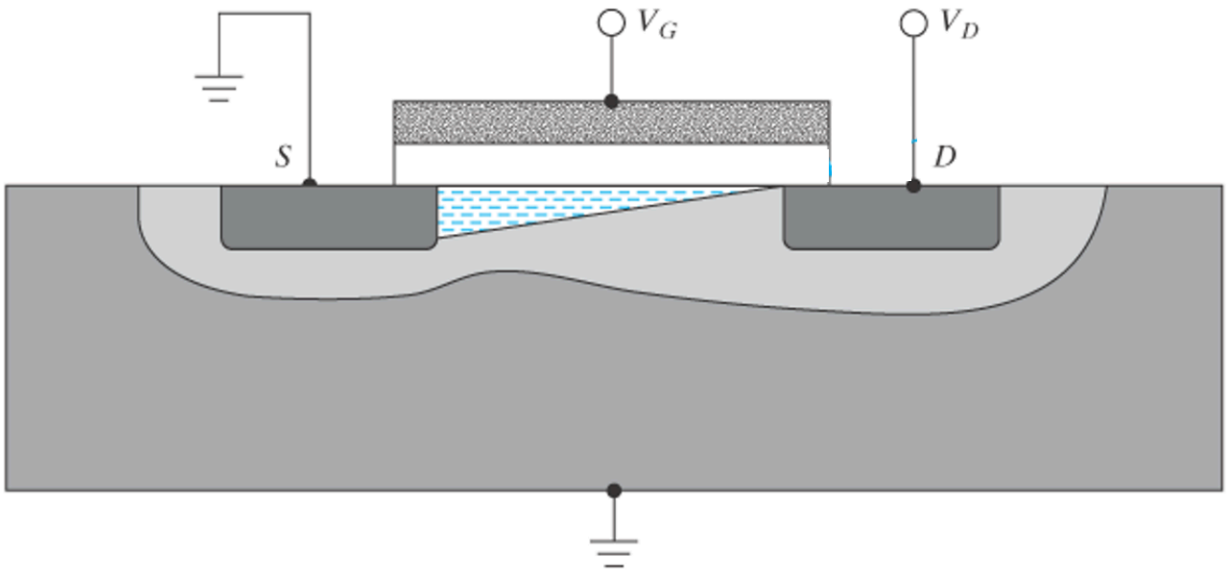
(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)

**Select one or more alternatives**

- [ ] The MOS capacitor is currently in thermal equilibrium ✔
- [ ] The semiconductor is doped with donor dopants
- [ ] To achieve flat band conditions we must apply a positive voltage across the MOS capacitor
- [ ] An inversion layer is currently present in the semiconductor ✔
- [ ] The work functions in the metal and semiconductor are identical
- [ ] The work functions in the metal is larger than that of the semiconductor
- [ ] The device is currently in strong inversion

Maximum marks: 4

**14**




The figure above shows the cross section of an enhancement mode (normally off) n-channel MOSFET at the onset of saturation.

**Based on the indicated depletion region and inversion layer, select the correct voltage relationships.**

*(correct: 2p, wrong: -2p, blank: 0p, minimum score: 0p)*

**Select one or more alternatives:**

- ☐ $V_D \approx 0\text{V}$
- ☐ $V_G \approx 0\text{V}$
- ☐ $V_D = V_G - V_T$ ✔
- ☐ $V_G < V_T$
- ☐ $V_G > V_T$ ✔

---

Maximum marks: 4

**15** The threshold voltage of a MOSFET is given by

$$V_T = \Phi_{ms} - \frac{Q_i}{C_i} - \frac{Q_d}{C_i} + 2\phi_F \quad (1)$$

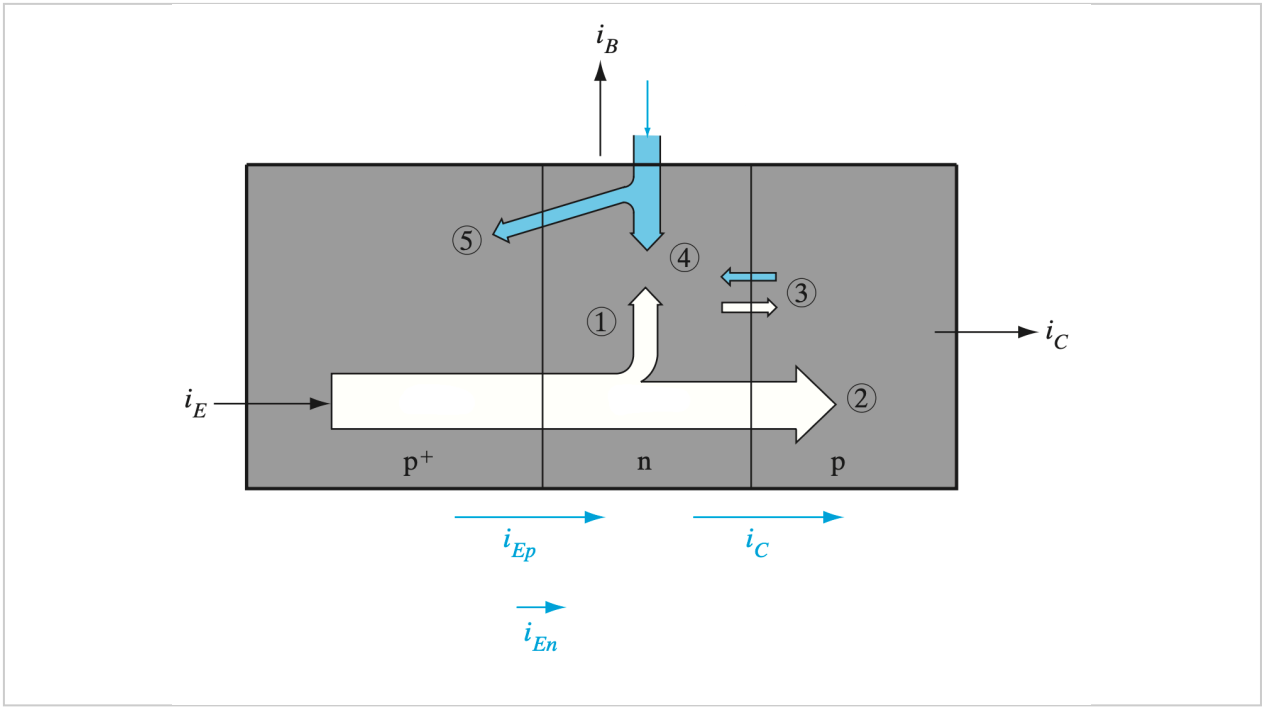
**Which of the terms in (1) will depend on the doping concentration in the substrate?**
*(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)*
**Select one or more alternatives:**

- ☐ $\Phi_{ms}$ ✔
- ☐ $-\frac{Q_d}{C_i}$ ✔
- ☐ $2\phi_F$ ✔
- ☐ $-\frac{Q_i}{C_i}$

---

Maximum marks: 6

**16**




*Fill in the blanks in the following section of text. Words are suggested for each blank – choose the most appropriate one. (correct: 1p, wrong: -1p, blank: 0p, minimum score: 0p)*

The figure shows a p-n-p bipolar transistor, biased in the normal active mode, which is usually

used for Select alternative (switching, rectification, amplification, power generation).

To have a good p-n-p transistor, we would prefer that almost all the holes injected by the emitter

are collected by the base. The n-type base region should therefore be Select alternative (heavily doped, narrow, wide, positively charged), and the hole lifetime should be

Select alternative (equal to the electron lifetime, long, short).

A second requirement is that the current $I_E$ crossing the emitter junction should be composed

almost entirely of Select alternative (electrons, holes, ions) injected to the base. This

requirement is satisfied by Select alternative (doping the base region lightly compared with the collector, doping the base region lightly compared with the emitter, doping the base region heavily compared with the emitter, doping the base region heavily compared with the collector).

In the figure, the hole and electron flows in the p-n-p bipolar transistor are labeled from 1-6:

(1) injected holes Select alternative (scattering, thermally generated, oscillating, lost to recombination) in the base

(2) holes reaching the Select alternative (forward, non, reverse)-biased collector junction

(3) thermally generated electrons and holes making up the Select alternative (injection current, reverse saturation current, hole current, base current) of the collector junction

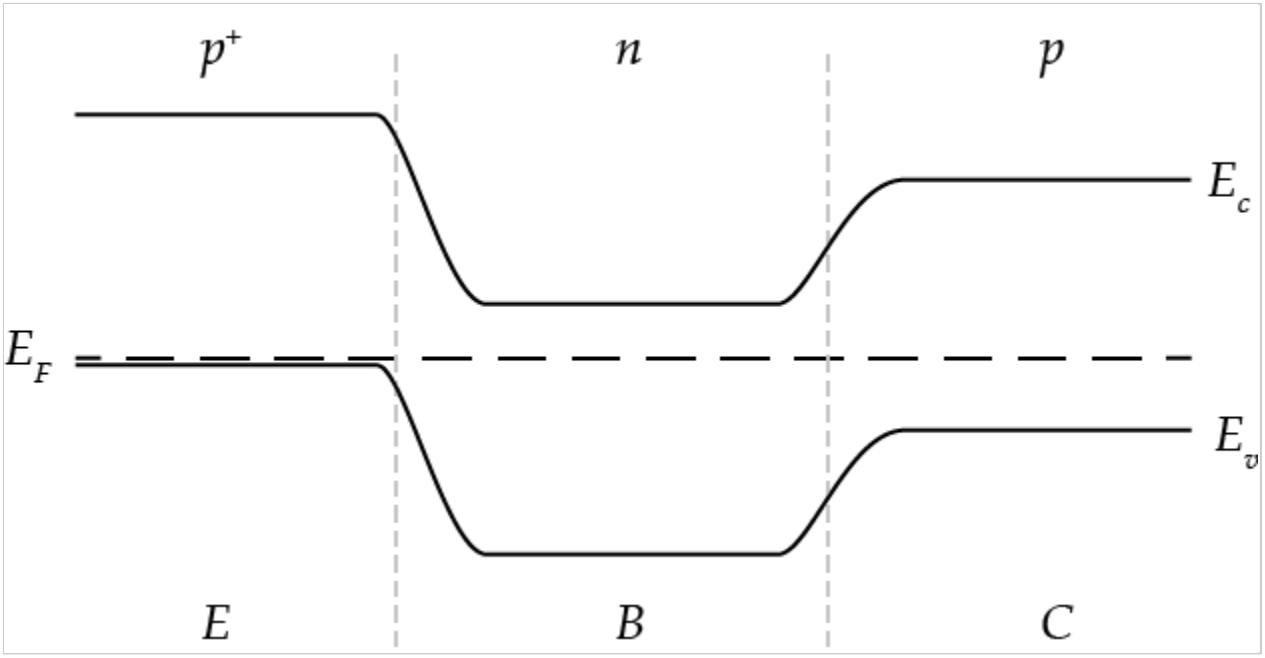
(4) electrons supplied by the base contact for Select alternative (thermal generation of charge carriers, stimulated emission, recombination with holes, doping the base region)

(5) electrons injected across the Select alternative (reverse, non, forward)-biased emitter junction

---

Maximum marks: 10

**17**




The illustration shows the simplified band diagram for a bipolar junction transistor (p-n-p). The emitter (E), base (B) and collector (C) regions are indicated. Which of the following statements are correct for the device with this band diagram?

(correct: 2p, wrong: -2p, blank: 0p, minimum score 0p)

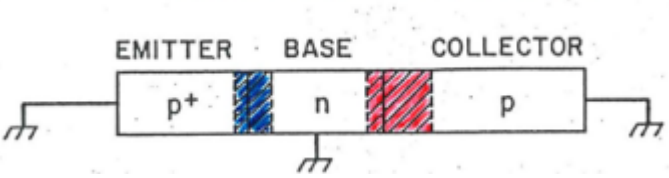
**Select one or more alternatives:**

- [ ] The device is currently operated in the normal active mode
- [ ] The dopant concentration is higher in the emitter than in the collector ✔
- [ ] There is currently a reverse bias across the B-C junction
- [ ] The device is currently in thermal equilibrium ✔
- [ ] The diffusion current across the E-B junction will be dominated by electrons

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Maximum marks: 4

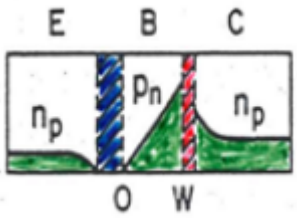
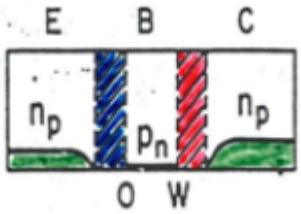
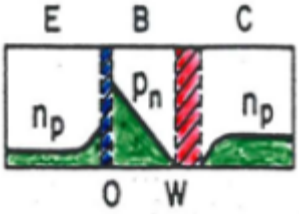
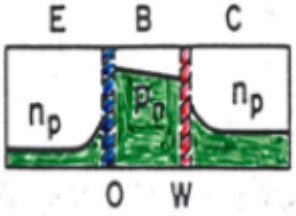
**18**




The figure above shows a p-n-p bipolar junction transistor in thermal equilibrium.
The figures below show the minority carrier distributions in the emitter (E), base (B), and collector (C), when the p-n-p transistor is operating in different regimes.

**Match the figures and the names of the different operation regimes.**
*(correct: 1p, wrong: 0p, blank: 0p, minimum score 0p)*

**Please match the values:**

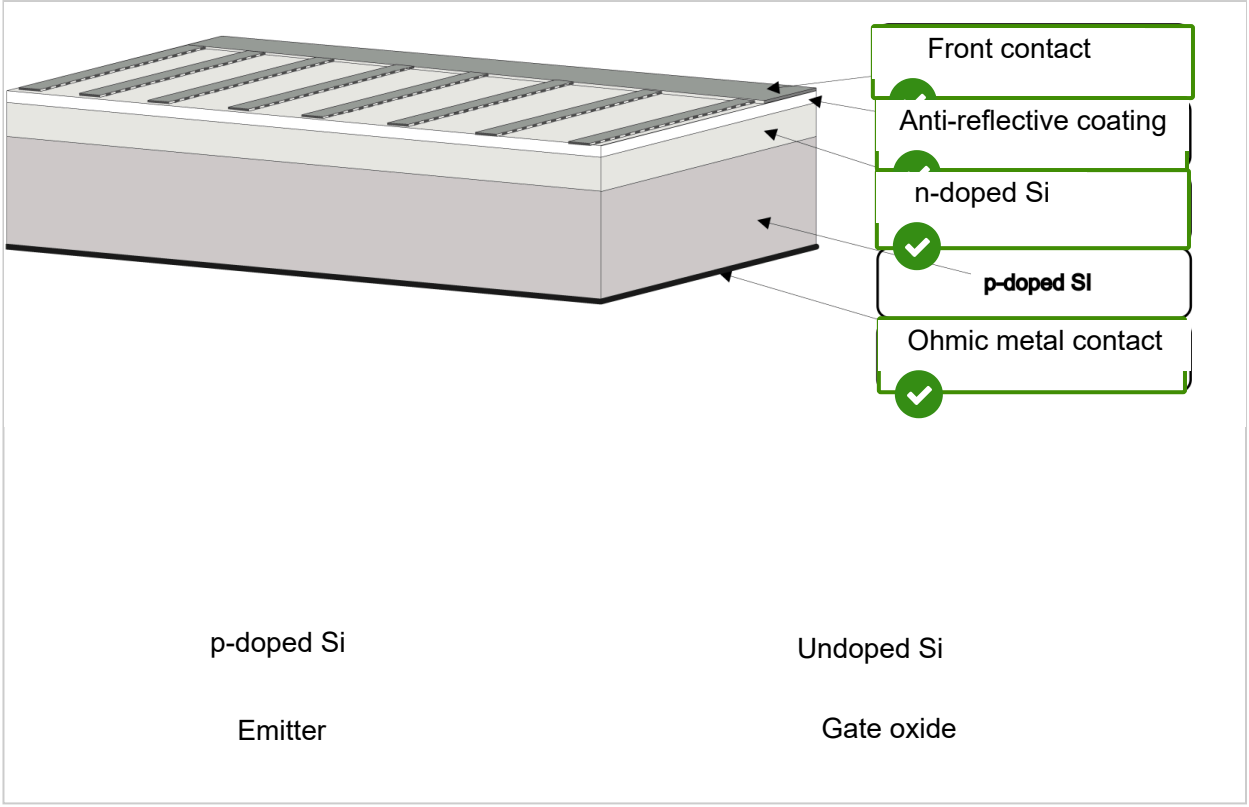
| | Cut-off | Normal active | Saturation | Inverted |
|---|---|---|---|---|
|  | ○ | ○ | ○ | ○ ✔ |
|  | ○ ✔ | ○ | ○ | ○ |
|  | ○ | ○ ✔ | ○ | ○ |
|  | ○ | ○ | ○ ✔ | ○ |



Maximum marks: 4

# Solar cell

The schematic below shows a common solar cell setup. Place the correct descriptions to the corresponding labels below.




p-doped Si

Undoped Si

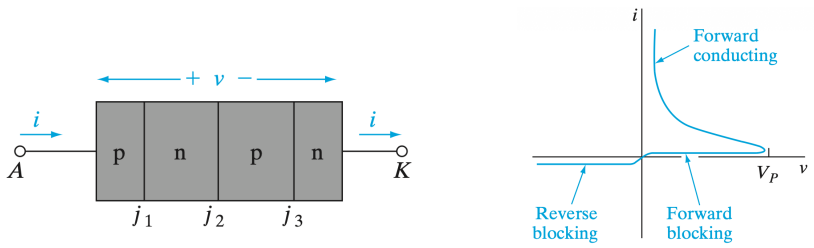
Emitter

Gate oxide

Correct. 4 of 4 marks. Try again

# The p-n-p-n diode

The p-n-p-n diode is a four-layer semiconductor device with three junctions. We connect the diode to two terminals, $A$ and $K$. The full diode is shown in Figure 1a. In the following you must make use of your knowledge about pn-junctions and BJTs to answer the questions about this device that you have probably not encountered before.

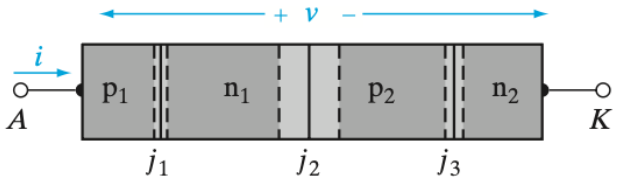



(a) (b)

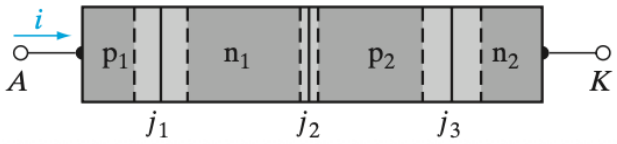
Figure 1: (a) The p-n-p-n diode. (b) I-V characteristics of the p-n-p-n diode.

**(a)** Sketch the depletion regions under (i) a small forward bias and (ii) reverse bias.

The depletion regions of the pnpn diode under a small forward bias and reverse bias are shown in Figure 2.




(a)




(b)

Figure 2: Depletion regions of the pnpn diode in the (a) forward-blocking and (b) reverse-blocking state.

Under these biasing conditions, the diode is in the *forward-blocking* and *reverse-blocking* state, respectively.

**(b)** Explain why the current passing through the device is limited to a very small value under both of the biasing conditions in (a). Assume that no carriers diffuse across the n and p areas

on each side of the $j_2$ junction.

In both of these cases we have both forward and reverse biased pn junctions. In the forward-blocking case, $j_1$ and $j_3$ are forward biased while $j_2$ is reverse biased. In the reverse-blocking case, we have the opposite.

In both cases, the current passing through the device is restricted to the small saturation current arising from thermal generation of carriers in the reverse-biased junctions.

When we apply a high enough voltage across the device (higher than the critical peak forward voltage $V_p$), the p-n-p-n diode switches to the *forward-conducting* state. We can then lower the voltage again, and the p-n-p-n diode will stay in the *forward-conducting state*, as shown in the I-V characteristic in Figure 1b. The device thus functions as a switch which is in the "off" state until switched by an external signal (a voltage pulse higher than $V_p$), and remains in the "on" state after switching.

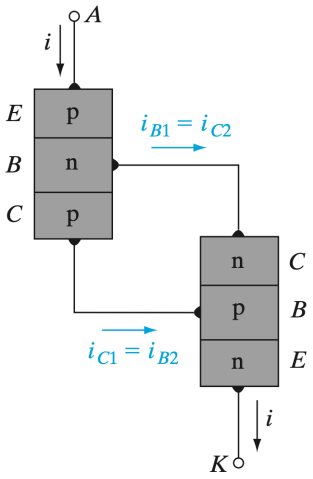



Figure 3: The two-transistor analogy of the p-n-p-n diode.

To understand this behavior, we can model the p-n-p-n diode as two connected bipolar junction transistors, a p-n-p and an n-p-n, as shown in Figure 3, where $j_1$ and $j_2$ form the emitter and collector junctions in the p-n-p transistor and $j_2$ and $j_3$ form the collector and the emitter junctions of the n-p-n transistor. We see that the collector current of the p-n-p transistor drives the base of the n-p-n, and oppositely that the collector current of the n-p-n drives the base of the p-n-p.

**(c)** When we increase the forward bias over the whole p-n-p-n device, what do you expect will happen to the depletion region of $j_2$? With the first transistor (the p-n-p) in mind, how will this affect $i_{C1}$?

When we increase the forward bias over the device, the depletion region width of the $j_2$ junction will increase.

If we first look at the first transistor, the pnp, an increase in the depletion region width of the base-collector junction will lead to the effective width of the base decreasing. This effect is known

as *base narrowing*. A decreased base width will allow more carriers to make it across the base region without being lost to recombination, increasing the collector current $i_{C1}$.

Note that a similar argument holds for $i_{C2}$ and the second (n-p-n) transistor.

**(d)** How does the change in $i_{C1}$ affect the second transistor (the n-p-n)? What do you expect will happen to the p-n-p-n device as a whole? *Hint: Note that the n-p-n transistor affects the p-n-p transistor as well!*

As the collector current of the first transistor $i_{C1}$ drives the second (n-p-n) transistor, an increase in $i_{C1}$ is an increase in $i_{B2}$. Increasing $i_{B2}$ increases the collector current of the second transistor, $i_{C2}$. This current again drives the first transistor, leading to an even larger $i_{C1}$.

At the same time, the second transistor affects the first in the same way.

The circle of increasing collector and base currents drives both of the transistors into saturation, and the reverse-biased $j_2$ becomes forward-biased. The p-n-p-n diode is now fully open, and we are in the "on" state.

**(e)** Why does the switch stay in its "on" state even after we lower the device voltage to a small forward bias?

In the "on" state, both transistors are in saturation and all three junctions become forward-biased. When we lower the voltage to a small forward bias, the two transistors continue to drive each other, stay in saturation and the diode as a whole stays in the "on" state.